

ONGUDAY, Russian Federation

WMO: 362310

Lat: 50.746N

Lon: 86.142E

Elev: 2733

StdP: 13.30

Time Zone: 7.00 (E07)

Period: 82-94

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.9	-23.4	-31.9	1.0	-27.4	-27.6	1.3	-22.3	11.5	14.4	7.9	19.3	0.0	N/A	0.589

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.9	82.0	62.2	78.5	60.9	75.3	59.8	65.3	75.6	63.2	74.5	61.5	72.4	2.0	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.1	92.2	68.9	59.4	83.6	66.4	57.2	77.3	65.5	31.9	75.6	30.1	75.0	28.8	72.4	78.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.3	8.6	6.4	-30.2	91.5	5.5	8.3	-34.2	97.5	-37.5	102.4	-40.6	107.0	-44.6	113.1
			-30.7	71.1	5.8	4.7	-34.9	74.4	-38.2	77.2	-41.5	79.8	-45.7	83.2
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	33.3	0.2	3.9	21.4	38.7	51.2	58.5	62.8	58.8	48.8	35.8	13.9	3.3
	DBStd	24.54	10.84	11.23	11.45	7.77	6.54	6.14	4.82	4.50	5.81	9.96	15.76	12.89
	HDD50	7207	1545	1290	886	343	67	5	0	2	90	447	1084	1448
	HDD65	11625	2010	1710	1351	788	427	207	97	197	486	905	1534	1913
	CDD50	1105	0	0	0	5	106	261	397	275	54	7	0	0
	CDD65	47	0	0	0	0	0	13	29	5	0	0	0	0
	CDH74	955	0	0	0	1	51	264	416	214	9	0	0	0
	CDH80	187	0	0	0	0	7	48	95	37	0	0	0	0
Wind	WSAvg	1.3	0.5	0.9	1.2	2.6	2.2	1.5	1.4	1.0	1.2	1.1	0.9	0.6
Precipitation	PrecAvg	14.3	0.2	0.2	0.2	0.7	1.7	2.6	2.8	2.5	1.5	0.8	0.4	0.4
	PrecMax	21.7	0.7	1.1	1.2	2.4	3.9	5.9	5.8	5.6	2.8	3.1	0.9	1.7
	PrecMin	7.5	0.0	0.0	0.0	0.0	0.4	1.1	0.9	0.6	0.5	0.0	0.0	0.0
	PrecStd	2.9	0.2	0.2	0.2	0.5	0.7	0.9	1.1	1.0	0.6	0.6	0.2	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	29.4	34.9	58.2	69.7	78.8	85.7	88.9	84.6	75.6	67.1	50.0	40.2
		MCWB	24.9	28.7	43.2	52.0	57.9	64.9	65.7	61.4	56.8	53.0	41.3	34.6
	2%	DB	25.4	29.3	48.7	65.5	74.1	80.4	82.7	79.9	70.8	60.6	43.0	30.9
		MCWB	21.8	24.1	38.3	49.1	55.6	62.3	62.6	60.9	54.4	49.7	36.0	27.2
	5%	DB	20.7	24.9	42.7	60.5	70.6	77.0	78.8	75.9	66.9	55.0	37.6	25.8
		MCWB	17.8	21.0	34.5	46.7	54.1	60.7	62.0	59.6	53.0	45.6	32.6	23.2
	10%	DB	16.4	20.5	37.8	56.1	66.9	72.3	75.0	72.0	62.8	49.8	33.4	21.5
		MCWB	14.4	17.5	31.7	44.2	52.3	58.9	61.1	58.7	51.3	42.7	28.6	19.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.9	29.1	43.8	54.5	59.7	68.8	69.7	65.4	61.4	54.3	42.0	33.7
		MCDB	28.9	35.1	58.0	67.1	76.8	80.4	76.8	74.7	68.2	63.8	48.6	39.4
	2%	WB	21.8	24.4	38.9	50.5	56.9	64.4	66.2	62.9	56.7	50.6	37.5	27.8
		MCDB	25.1	29.0	47.8	63.3	71.4	76.0	75.9	75.8	66.6	59.3	42.4	30.3
	5%	WB	17.9	20.9	35.1	47.4	55.1	62.1	63.9	61.1	54.0	46.7	32.6	23.1
		MCDB	20.5	24.7	41.8	58.9	68.6	73.4	74.3	72.9	64.0	53.7	37.2	26.1
	10%	WB	14.4	17.6	32.0	44.4	53.1	60.1	62.2	59.5	51.7	43.3	28.7	19.2
		MCDB	16.4	20.4	37.5	55.0	64.8	70.5	73.0	69.9	61.5	49.2	32.9	21.5
Mean Daily Temperature Range	5% DB	MDBR	16.6	21.7	23.7	26.1	24.4	22.6	21.9	23.5	23.9	21.2	16.5	15.5
		MCDBR	18.3	22.7	28.1	35.9	33.6	31.3	30.1	33.8	33.7	29.8	19.3	20.5
		MCWBR	15.6	18.3	18.5	21.7	18.4	15.5	13.4	16.5	19.1	19.4	14.1	16.4
	5% WB	MCDBR	18.3	21.9	26.9	32.1	30.8	27.8	25.0	28.4	29.7	28.1	18.1	18.7
		MCWBR	15.8	18.0	18.2	20.6	18.3	14.2	13.9	16.1	19.0	19.0	13.9	15.9
Clear-Sky Solar Irradiance	taub	0.258	0.288	0.341	0.394	0.385	0.386	0.398	0.377	0.337	0.324	0.287	0.259	
	taud	2.179	2.132	2.062	2.125	2.250	2.338	2.321	2.352	2.417	2.343	2.232	2.147	
	Ebn at Noon	243	264	268	267	276	276	270	271	271	250	227	220	
	Edh at Noon	23	32	42	44	41	38	38	35	30	26	22	20	
All-Sky Solar Radiation	RadAvg	443	789	1265	1594	1769	1878	1765	1585	1196	708	423	325	
	RadStd	36	44	78	99	150	167	98	111	103	80	36	31	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page